M4. Weekly Circulation Report

Category B Practice - Problem 4 (Intermediate)

Approach

The report uses one StringBuilder and displayInfo() polymorphically. A guarded instanceof check makes the StudentMember downcast safe, allowing the course to be appended only for student records.

Java Solution

```
class LibraryMember {
    protected final String memberId; protected final int borrowLimit; protected int booksBorrowed;
    public LibraryMember(String id,int limit) {
        if(id==null||id.trim().length()<4) throw new IllegalArgumentException("Invalid ID");
        if(limit<=0) throw new IllegalArgumentException("Limit must be positive");
        memberId=id; borrowLimit=limit;
    }
    public String displayInfo() { return "General | Books: " + booksBorrowed; }
}
class StudentMember extends LibraryMember {
    private final String course;
    public StudentMember(String id,int limit,String course) { super(id,limit); this.course=course; }
    public String getCourse() { return course; }
    @Override public String displayInfo() { return "Student | Course: " + course + " | Books: " + booksBorrowed; }
}
class WeeklyReport {
    static String batchPrint(LibraryMember[] members) {
        StringBuilder report=new StringBuilder();
        for(LibraryMember member:members) {
            if(report.length()>0) report.append(" | ");
            report.append(member.displayInfo());
            if(member instanceof StudentMember) {
                StudentMember student=(StudentMember)member;
                report.append(" [Course via downcast: ").append(student.getCourse()).append("]");
            }
        }
        return report.toString();
    }
}
```

Sample Output

```
General | Books: 0 | Student | Course: ECE | Books: 0 [Course via downcast: ECE] |
```